

# Sampling

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210 Psychological Methods



# Road Map

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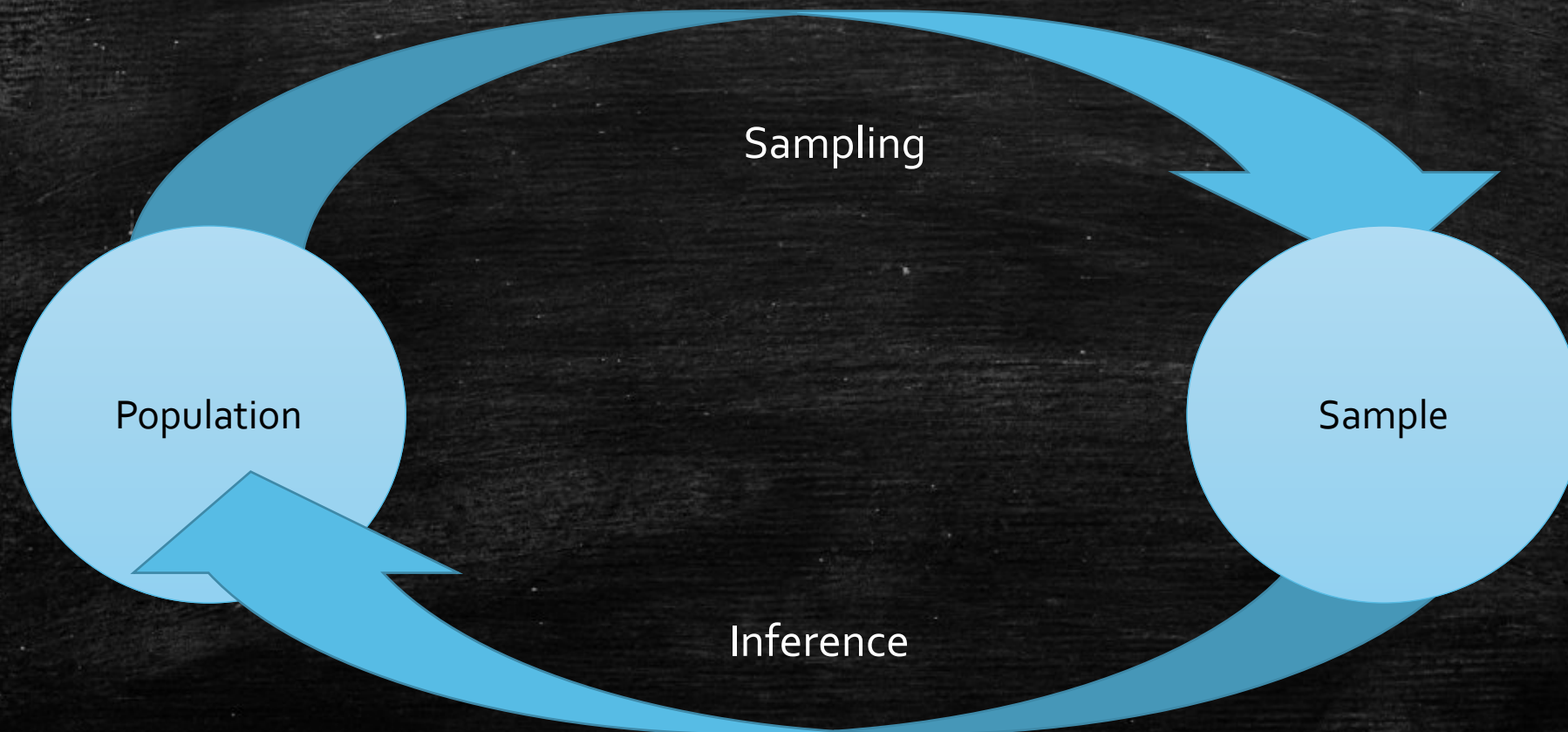
- Sampling
  - Ann Landers Example
- How can Sampling Go Wrong?



# Sampling

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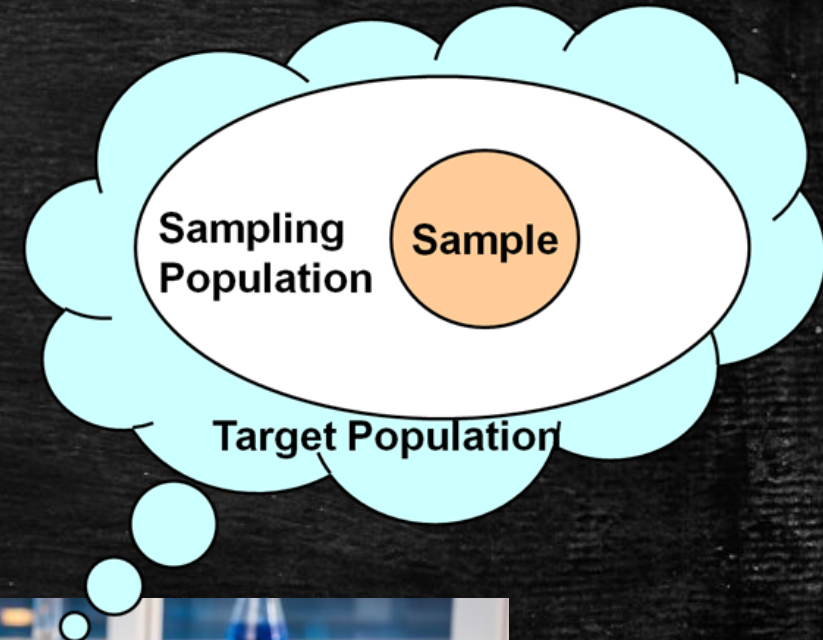
- Third Step in Study
  - Collect units to study





# Definitions

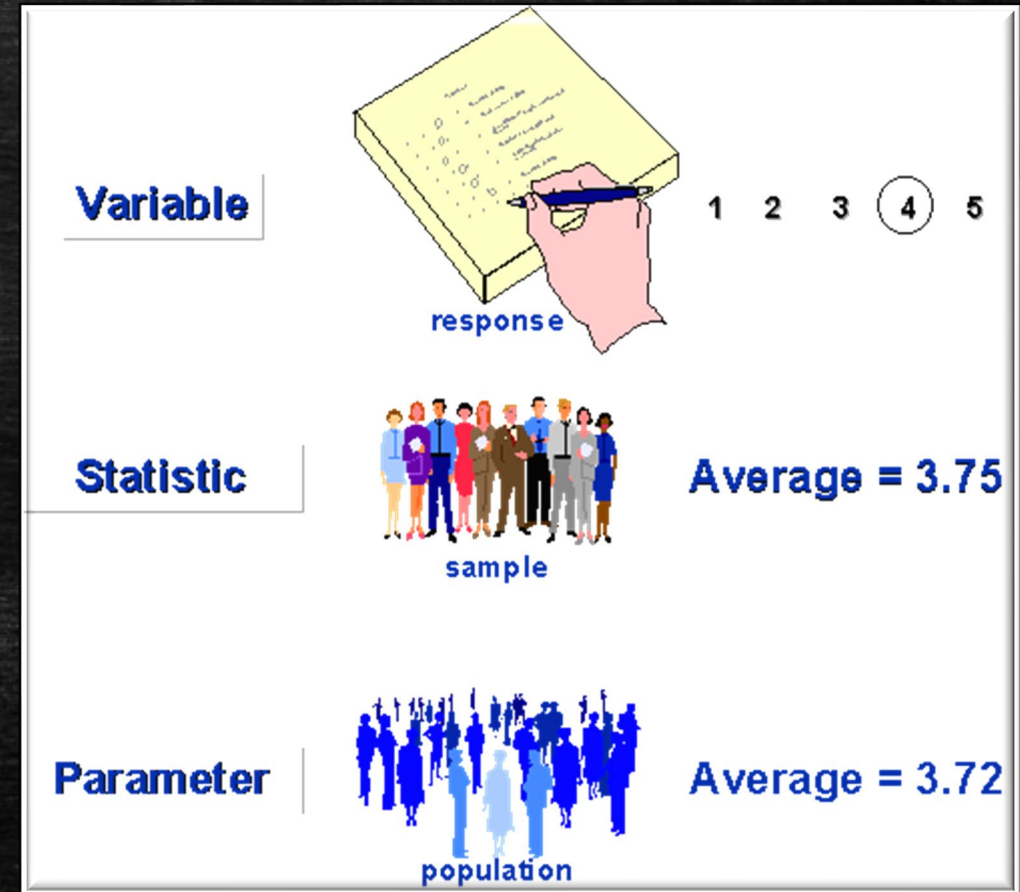
- Population
  - entire group about whom we want information
- Sample
  - part of a population from which we collect information
- Sampling frame
  - the list of units from which samples are chosen





# Definitions

- **Parameter**
  - number that is associated with a population
- **Statistic**
  - a number that describes a sample
  - We use statistics to estimate parameters
- We draw inferences from a sample to the population



**Variable**



1 2 3 4 5

**Statistic**



**Average = 3.75**

**Parameter**



**Average = 3.72**



# Representative sample

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- We want a sample to represent a population
  - If not, how does the sample differ from the population?
- Sampling bias leads to systematic errors
  - how a sample represents the population



## ACTIVITY: Ann Landers (Dear Ann)

- In 1975, Dear Ann asked her readers...
- "If you had it to do over again – would you have children?"
  - 15,000 postcards
  - 70% said No

**DEAR ABBY:**

### ***My Dad Objects To a Pet Monkey***

**D**EAR ABBY: I am 10 years old and my Daddy said that when I saved enough money I could buy anything I wanted with it.

All my life, I have wanted a monkey. I have saved \$14. I asked Daddy if I could buy a pet monkey and he said no, because I wouldn't know how to take care of it. My Mom is the fussy type. You know, everything has to be just so. Do you know anyone who has a pet monkey, and can give me some advice?

**WANTS A MONKEY**

**DEAR WANTS:** I have had two pet monkeys (David and Bathsheba) and, although I love monkeys, your father is right. To quote my son (he was 4 at the time), "Monkeys should live with monkeys, and people should live with people."



**ABIGAIL  
VAN BUREN**



## ACTIVITY: Ann Landers (Dear Ann)

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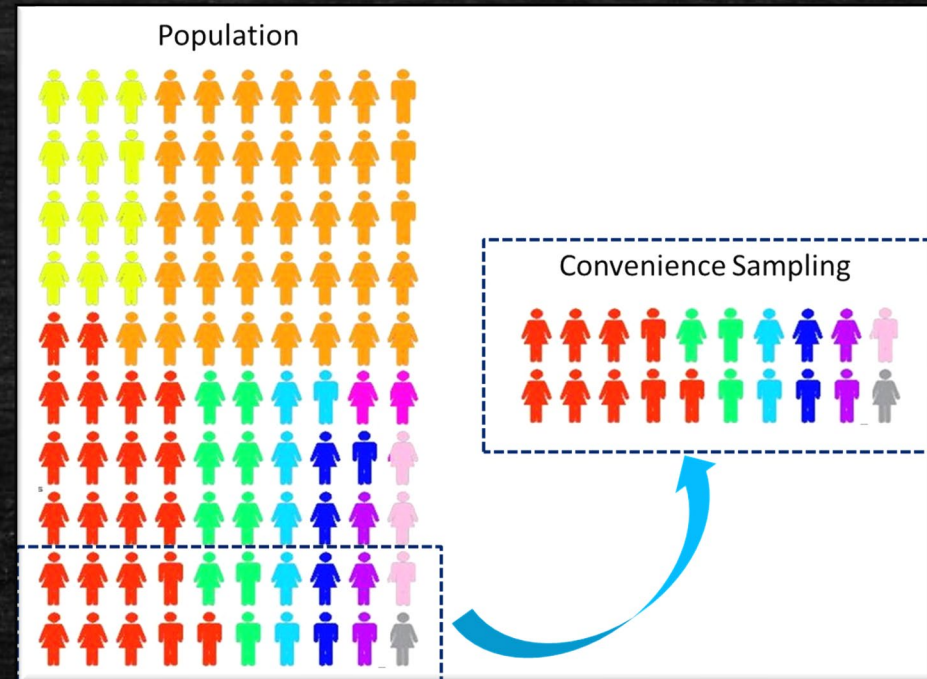
- Good Housekeeping asked its readers, after the results came in...
- "If you had it to do over again – would you have children?"
  - 95% said Yes
- How is this possible?
  - What kind of bias could explain these differences?





# Convenience sample

- Convenience sample
  - Use an available but non-representative set of respondents
  - Examples
    - In the Classroom
    - Research Driven Sampling
    - Growing Trends
    - Biological Surveys





# Convenience sample

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- Convenience sampling can be effective in a wide variety of situations,
  - simply depending on what type of data the researcher wishes to collect.
- Only the researcher can decide whether this type of sampling is worthwhile to their data collection efforts.



# Random Sampling

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- Best way to ensure a representative sample
- Probability sample
  - a sample in which each element has a known (a priori) probability of being drawn
- Simple random sample (SRS)
  - every sample of a given size has an equal probability of being chosen.
- Random mechanisms
  - Random digit table (*Rand Corporation 1955*)
  - Computer generated random numbers (used to be call pseudorandom)



## TABLE OF RANDOM DIGITS

1

|       |       |       |       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 00000 | 10097 | 32533 | 76520 | 13586 | 34673 | 54876 | 80959 | 09117 | 39292 | 74945 |
| 00001 | 37542 | 04805 | 64894 | 74296 | 24805 | 24037 | 20636 | 10402 | 00822 | 91665 |
| 00002 | 08422 | 68953 | 19645 | 09303 | 23209 | 02560 | 15953 | 34764 | 35080 | 33606 |
| 00003 | 99019 | 02529 | 09376 | 70715 | 38311 | 31165 | 88676 | 74397 | 04436 | 27659 |
| 00004 | 12807 | 99970 | 80157 | 36147 | 64032 | 36653 | 98951 | 16877 | 12171 | 76833 |
| 00005 | 66065 | 74717 | 34072 | 76850 | 36697 | 36170 | 65813 | 39885 | 11199 | 29170 |
| 00006 | 31060 | 10805 | 45571 | 82406 | 35303 | 42614 | 86799 | 07439 | 23403 | 09732 |
| 00007 | 85269 | 77602 | 02051 | 65692 | 68665 | 74818 | 73053 | 85247 | 18623 | 88579 |
| 00008 | 63573 | 32135 | 05325 | 47048 | 90553 | 57548 | 28468 | 28709 | 83491 | 25624 |
| 00009 | 73796 | 45753 | 03529 | 64778 | 35808 | 34282 | 60935 | 20344 | 35273 | 88435 |
| 00010 | 98520 | 17767 | 14905 | 68607 | 22109 | 40558 | 60970 | 93433 | 50500 | 73998 |
| 00011 | 11805 | 05431 | 39808 | 27732 | 50725 | 68248 | 29405 | 24201 | 52775 | 67851 |
| 00012 | 83452 | 99634 | 06288 | 98083 | 13746 | 70078 | 18475 | 40610 | 68711 | 77817 |
| 00013 | 88685 | 40200 | 86507 | 58401 | 36766 | 67951 | 90364 | 76493 | 29609 | 11062 |
| 00014 | 99594 | 67348 | 87517 | 64969 | 91826 | 08928 | 93785 | 61368 | 23478 | 34113 |
| 00015 | 65481 | 17674 | 17468 | 50950 | 58047 | 76974 | 73039 | 57186 | 40218 | 16544 |
| 00016 | 80124 | 35635 | 17727 | 08015 | 45318 | 22374 | 21115 | 78253 | 14385 | 53763 |
| 00017 | 74350 | 99817 | 77402 | 77214 | 43236 | 00210 | 45521 | 64237 | 96286 | 02655 |
| 00018 | 69916 | 26803 | 66252 | 29148 | 36936 | 87203 | 76621 | 13990 | 94400 | 56418 |
| 00019 | 09893 | 20505 | 14225 | 68514 | 46427 | 56788 | 96297 | 78822 | 54382 | 14598 |
| 00020 | 91499 | 14523 | 68479 | 27686 | 46162 | 83554 | 94750 | 89923 | 37089 | 20048 |
| 00021 | 80336 | 94598 | 26940 | 36858 | 70297 | 34135 | 53140 | 33340 | 42050 | 82341 |
| 00022 | 44104 | 81949 | 85157 | 47954 | 32979 | 26575 | 57600 | 40881 | 22222 | 06413 |
| 00023 | 12550 | 73742 | 11100 | 02040 | 12860 | 74697 | 96644 | 89439 | 28707 | 25815 |



# No one random sample is perfectly representative.

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- They are representative **on average**
- We can expect the pattern across samples -> sampling distribution
- The bigger the sample, the closer it is to the population, the more representative
  - When sample = population , census
  - Sampling fraction
    - $n$  denotes sample size
    - $N$  denotes population size
    - Sampling fraction =  $n/N$



# Other Sampling designs

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- Multistage samples
  - Also called cluster samples
  - We don't have a sampling frame of units of interest
  - We sample higher level units that do have sampling frames
  - Example
    - Sample 5 of 50 states
    - Samples 100 census tracts in each state
    - Sample 10 people from each census tract
    - $n=5,000$



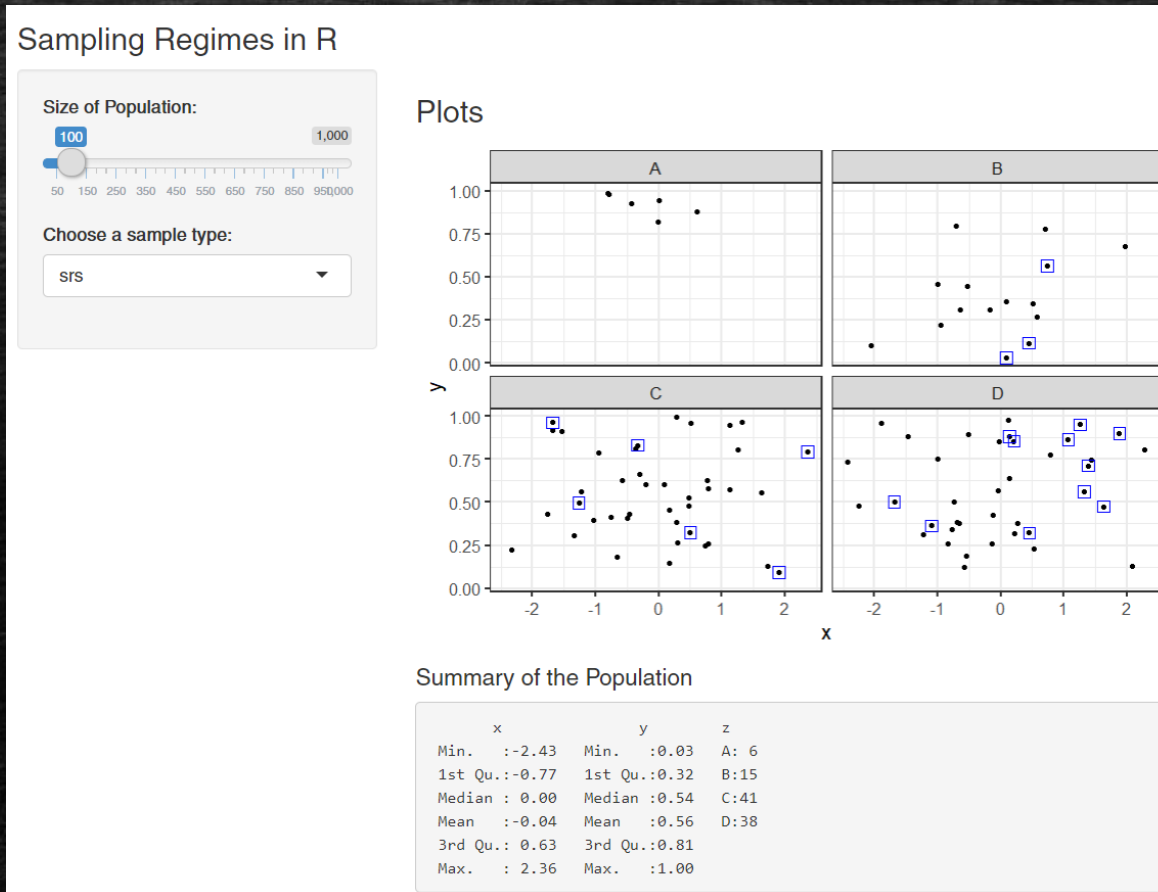
# Other Sampling designs

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- Stratified random sampling
  - Goal: to increase precision of estimates that come from sample
  - Mechanism – use strata in which responses are more similar to one another than average
- We can use multistage sampling AND strata random sampling in the same study



# Illustration in R





Wrapping Up...

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How can Sampling Go Wrong?

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# Unintentional Bias

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- Assumptions from the Researcher
  - Researchers have to make judgements
    - who to sample, how to sample, and how close to the population those samples are
  - Major Challenge in Political Polling
    - US Election of 1948
    - UK Snap Election 2017



# Election of 1948

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- Use of Quota sampling
  - introduced by George Gallup as early as 1935
  - successfully used by him to predict the winner of the 1936, 1940 and 1944 elections.
- Quota sampling is a systematic effort to
  - force the sample to fit a certain national profile by using quotas
- The sample should have
  - so many women, so many men, etc
  - represent the same proportions in the sample as are in the electorate at large.



# 1948 Presidential Election



## United States presidential election, 1948



– 1944

November 2, 1948

1952 –

All 531 electoral votes of the [Electoral College](#)  
266 electoral votes needed to win

Turnout

53.0%<sup>[1]</sup> ▼ 2.9 pp



Nominee

Harry S. Truman

Thomas E. Dewey

Strom Thurmond

Party

Democratic

Republican

Dixiecrat

Home state

Missouri

New York

South Carolina

Running mate

Alben W. Barkley

Earl Warren

Fielding L. Wright



# Election of 1948

- For the 1948 election, between
  - Thomas Dewey and Harry Truman,
  - Gallup conducted a poll with a
    - sample size of about 3250.
- Each individual in the sample was interviewed in person
  - by a professional interviewer
    - to minimize nonresponse bias,
  - and each interviewer was given
    - a very detailed set of quotas to meet.





# Election of 1948

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- For example, an interviewer could have been given the following quotas:
  - seven white men under 40 living in a rural area,
  - five black men under 40 living in a rural area,
  - six black women under 40 living in a rural area, etc.
- Other than meeting these quotas the
  - ultimate choice of who was interviewed was left to each interviewer.



# Election of 1948

- Based on the results of this poll,
- Gallup predicted a victory for Dewey,
  - the Republican candidate.
- The predicted breakdown of the vote was
  - 50% for Dewey,
  - 44% for Truman, and
  - 6% for third-party candidates Strom Thurmond and Henry Wallace.

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# “Dewey Beats Truman” Election of 1948

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- On election night, an early press deadline required the first post-election issue of the *Tribune* to go to press
  - before even the East Coast states had reported many results from the polling places.
- The paper relied on its veteran Washington correspondent and political analyst Arthur Sears Henning, who had
  - predicted the winner in four out of five presidential contests in the past 20 years.
- Conventional wisdom, supported by polls, was almost unanimous that a Dewey presidency was "inevitable"





# Chicago Daily Tribune

THE WORLD'S GREATEST NEWSPAPER

HOME

VOL. CVII - NO. 264

(REG. U.S. PAT. OFFICE, COPYRIGHT 1948 BY THE CHICAGO TRIBUNE)

WEDNESDAY, NOVEMBER 3, 1948

FOUR CENTS - PAY NO MORE

## DEWEY DEFEATS TRUMAN

### G. O. P. Sweep Indicated in State; Boyle Leads in City

#### REPUBLICAN TICKET AHEAD OF 1944 VOTE

#### Town Balloting Gives Trend

##### Probable Winners

SENATOR—Wayland Brooks (R.)  
GOVERNOR—Dwight H. Green (R.)  
LT. GOV.—Richard Yates Rowe (R.)  
SECRETARY OF STATE—William G. Stratton (R.)  
AUDITOR—Simon A. Murray (R.)  
TREASURER—Elmer H. Droste (R.)  
ATTY. GEN.—George F. Barrett (R.)  
\*Re-elected.

##### BY ARTHUR EVANS

Early returns last night indicated that the Dewey-Warren Republican ticket and the state ticket headed by Sen. Brooks and

#### Tops Coghlan in Hot Race for Attorney

##### Probable Winners

STATE'S ATTORNEY—John S. Boyle (D.) or Malachy J. Coghlan (R.)  
RECORDER OF DEEDS—Victor L. Schlaeger (D.)  
CIRCUIT COURT CLERK—John E. Conroy (D.) or Mrs. Mabel Reinecke (R.)  
CORONER—A. L. Brodie (D.) or Frank Kalleux (R.)  
SUPERIOR COURT CLERK—Henry Soanenschein (D.)  
SANITARY DISTRICT TRUSTEES (3)—Frank W. Chesrow (D.), John A. Cullerton (D.), and Casimir Griglik (D.) or Walter E. McCarron (R.)  
MUNICIPAL COURT BAILIFF—Albert J. Moran (D.)  
MUNICIPAL COURT CLERK—Joseph L. Gill (D.)  
\*Re-elected.

##### BY GEORGE TAGGE

Heavy Democratic leads in Chicago wards darkened the outlook last night for Republican candidates for state's attorney and other Cook county offices.

#### RECORD CITY VOTE SEEN IN LATE TALLIES

#### Suburban Ballot Near 375,000

A high record vote seemed certain for Chicago yesterday in the first post-war Presidential election. A turnout of 1,900,000 was indicated in the final sampling of precincts in each of the 50 wards, according to Chief Clerk John S. Rusch of the Chicago election board. This prediction was based on a projection of 1,789,776 one hour before the polls closed.

TOPS 1944 RECORD  
The old record was established in 1944.

#### BULLETINS ON ELECTIONS

##### COOK COUNTY

President—749 pcts. of 5023 in Cook county: Truman D 182,780, Dewey R 128,916, Wallace P 575.

Senator—481 pcts. of 5023 in Cook county: Douglas D 128,596, Brooks R 70,465.

Governor—481 pcts. of 5023 in Cook county: Stevenson D 133,374, Green R 68,973.

Lieutenant Governor—14 pcts. of Cook county: Dixon D 3,968, Rowe R 1,495.

State's Attorney—524 pcts. of 5023 in Cook county: Boyle D 134,682, Coghlan R80,100, Heller P 3,959.

Secretary of State—14 pcts. of 4143 in Chicago: Barrett D 4,003, Stratton R 1,494.

State Treasurer—14 pcts. of 4,143 in Chicago: Smith D 3,994, Droste R 1,475.

Secretary of State—4 precincts of 9,231 in Illinois: Barrett D 415, Stratton R 633.

##### NATION

Birmingham, Ala., Nov. 2 (AP)—Gov. Thurmond, the State Rights Democratic candidate, won Alabama's 11 electoral votes.

Alabama 2 boxes of 2,408, Thurmond 116, Dewey 37, Wallace 3.

Arkansas—2 districts of 2,217 Truman 85; Dewey 2, Wallace 2.

Florida—60 precincts of 1,523, Dewey 8,983, Thurmond 3,173, Truman 11,657, Wallace 1,151; governor, 39 precincts, Acker R 2,369, Warren D 9,743.

Georgia—105 out of 1,736 precincts Truman 2,272, Dewey 325, Thurmond 690, Wallace

Ohio—1,575 polling places of 9,710, Truman 187,830, Dewey 221,233, Wallace 1,389; governor, 32 polling places Herbert R 3,264, Lausche D 2,978.

Oklahoma—senator 2 precincts of 3701 Kerr D 164 Risley R 89.

Rhode Island—4 districts of 266, Truman 545, Dewey 1,294, Wallace 9.

Rhode Island senator, 4 districts of 266; Green D 557, Hazard R 1,237.

Columbia, S.C., Nov. 2 (AP)—Gov. J. Strom Thurmond, States' rights presidential candidate, captured his home state's eight electoral college votes today.

#### Early Count Gives G.O.P. Senate Edge

##### BY WALTER TROHAN

Republicans took an early lead in one of the hottest campaigns in history for control of the senate. G.O.P. retention of control of the senate seemed assured when New Mexico appeared to be deserting solid Democratic rule.

Patrick J. Hurley, Republican, took an early lead over former Agriculture Secretary Anderson in New Mexico, one of eight pivotal states. This offset what was expected to be a certain Democratic gain in Oklahoma, where former Gov. Kerr, Democrat, leaped ahead of Rep. Risley (R.). The G.O.P. was also ahead in Kentucky where Sen. Cooper

#### PUTS G.O.P. BACK IN THE WHITE HOUSE

#### Sizable Electoral Margin Seen

##### BY ARTHUR SEARS HENNING

Dewey and Warren won a sweeping victory in the Presidential election yesterday.

The early returns showed the Republican ticket leading Truman 297 to 270 electoral votes. The early returns would disclose that Dewey won the Presidency by an overwhelming majority of electoral votes.

Herbert Brownell, manager of the Dewey





# Chicago Daily Tribune

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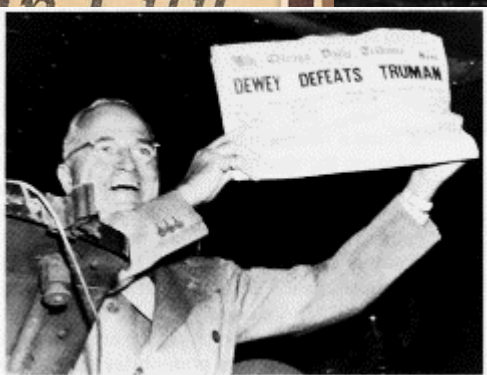
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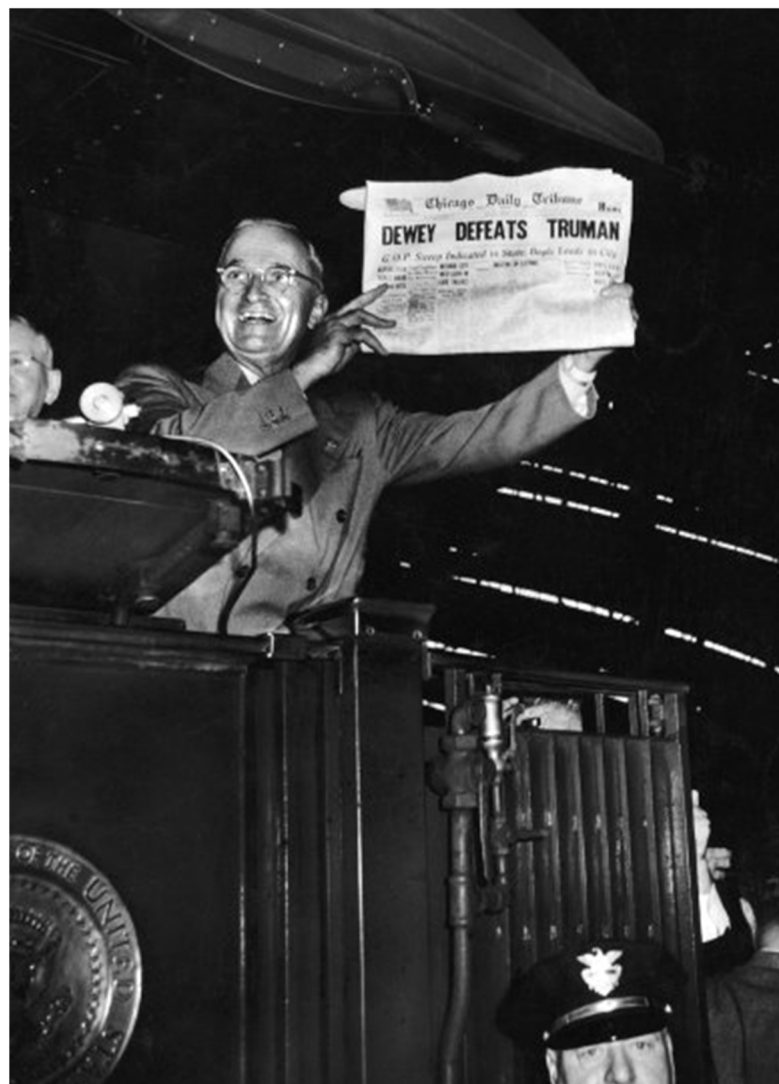


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# “Dewey Beats Truman”

- The actual results were almost exactly reversed:
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  - 5% for third-party candidates.

## United States presidential election, 1948



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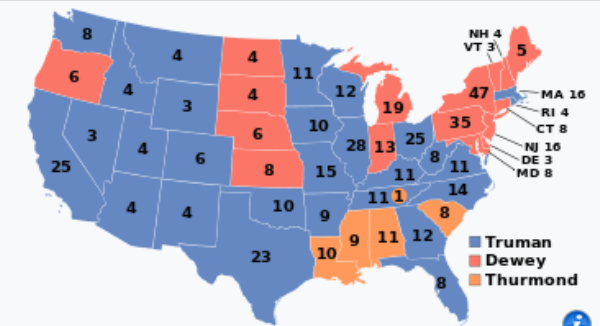
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| Nominee        | Harry S. Truman  | Thomas E. Dewey | Strom Thurmond     |
|----------------|------------------|-----------------|--------------------|
| Party          | Democratic       | Republican      | Dixiecrat          |
| Home state     | Missouri         | New York        | South Carolina     |
| Running mate   | Alben W. Barkley | Earl Warren     | Fielding L. Wright |
| Electoral vote | 303              | 189             | 39                 |
| States carried | 28               | 16              | 4                  |
| Popular vote   | 24,179,347       | 21,991,292      | 1,175,930          |
| Percentage     | 49.6%            | 45.1%           | 2.4%               |

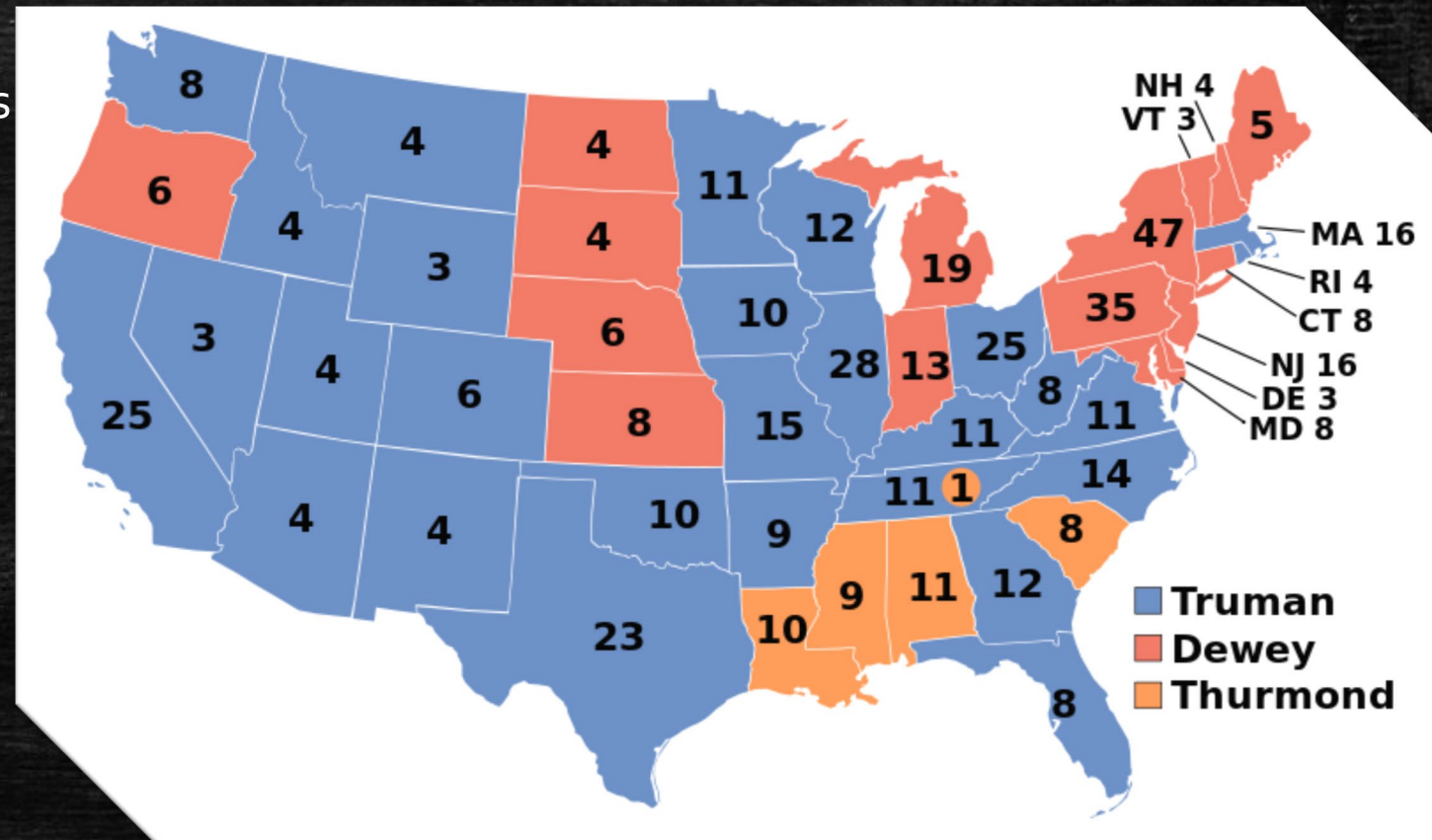


Presidential election results map. **Blue** denotes states won by Truman/Barkley, **red** denotes those won by Dewey/Warren, **orange** denotes those won by Thurmond/Wright, including a [Tennessee faithless elector](#). Numbers indicate the number of electoral votes allotted to each state.



# “Dewey Beats Truman” Election of 1948

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  - 50% for Truman,
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# UK Snap Election 2017

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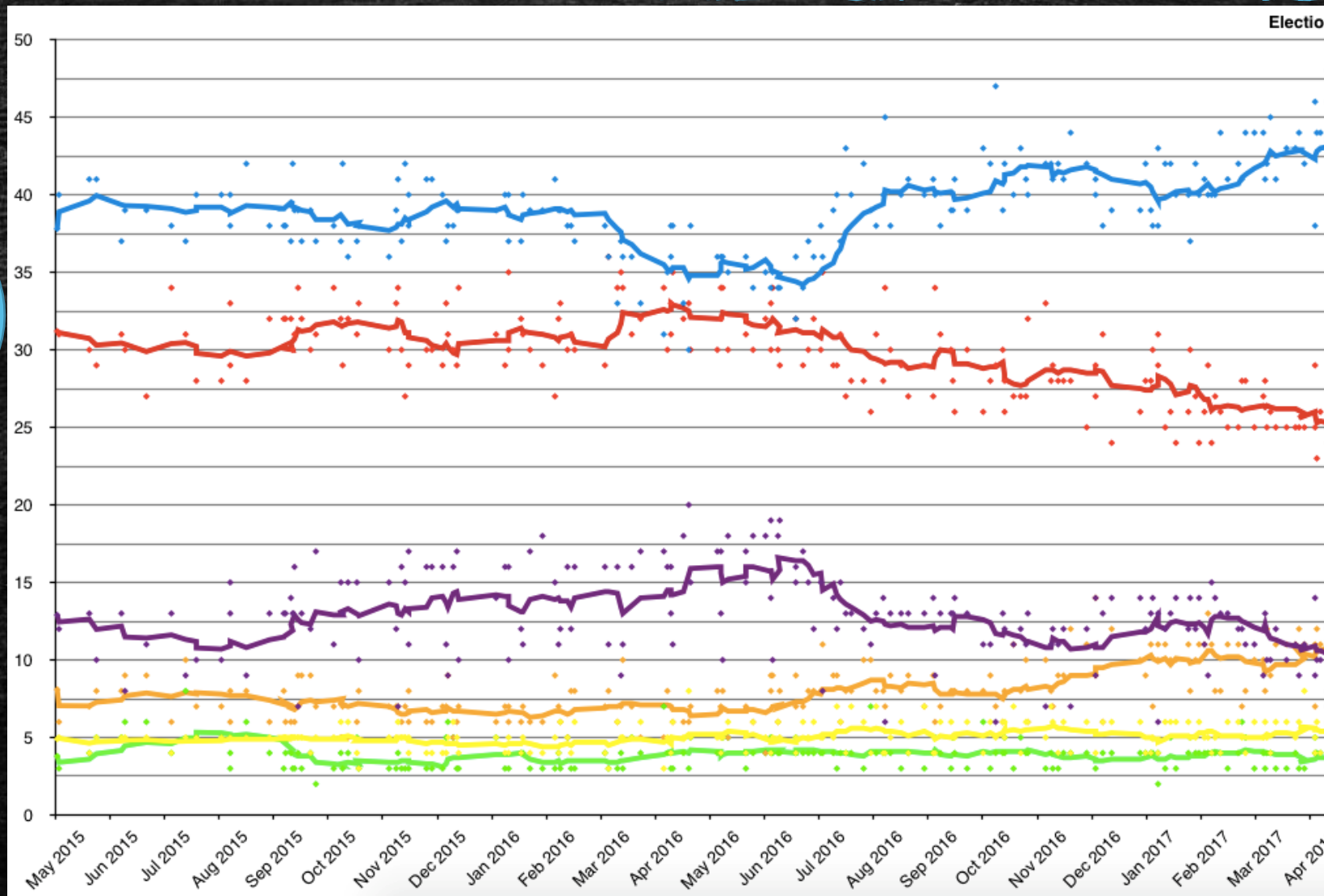
- Prime Minister Theresa May called for a snap election April 18
  - 3 years early (7 May 2020)
- May hoped to secure a larger majority for the Conservative Party
  - in order to "strengthen [her] hand in [the forthcoming Brexit] negotiations".
- She had 330 seats out of 650 in the house of commons
  - 325 are needed for a majority



# UK Snap Election 2017

Election  
Called

Last  
Election

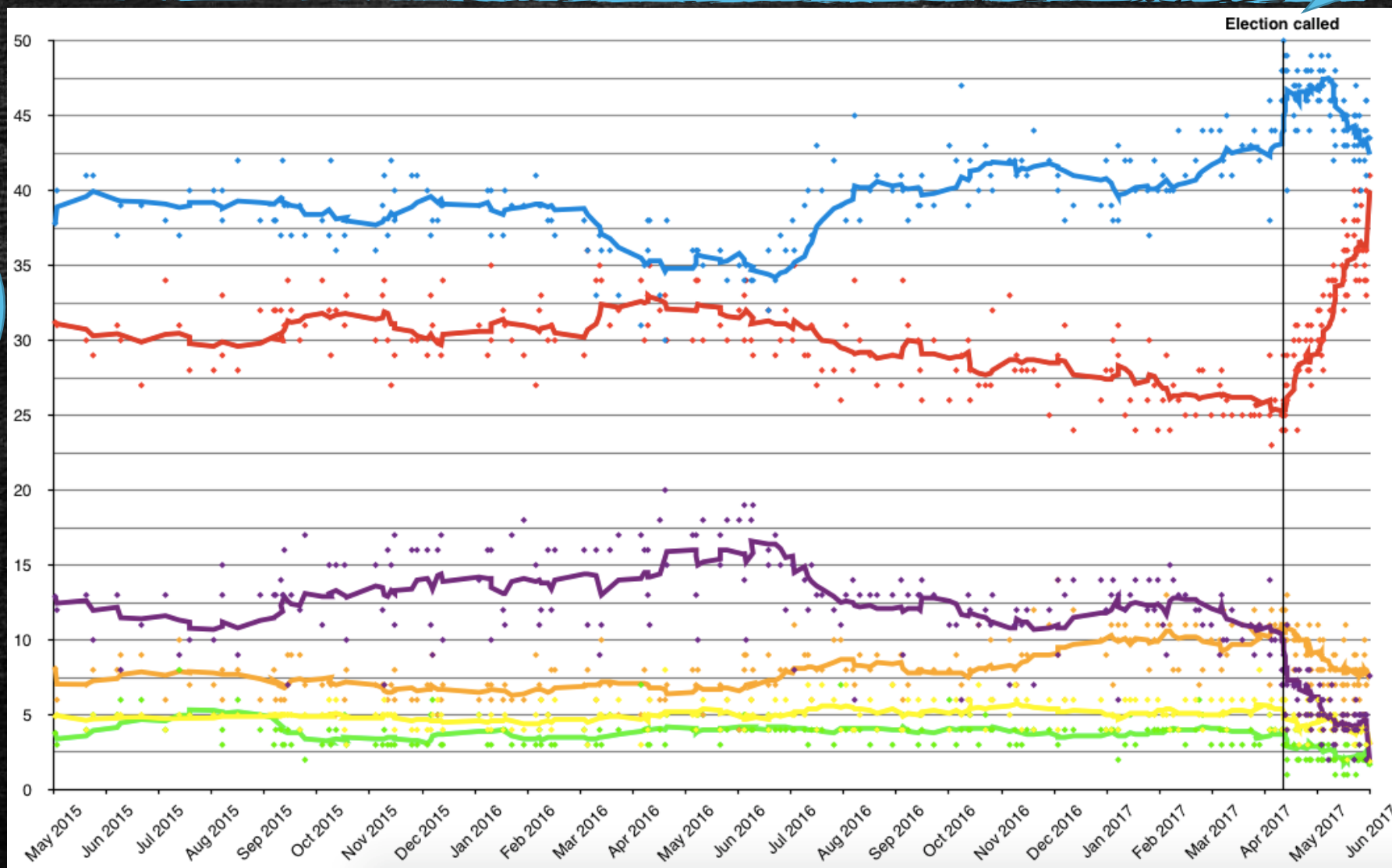




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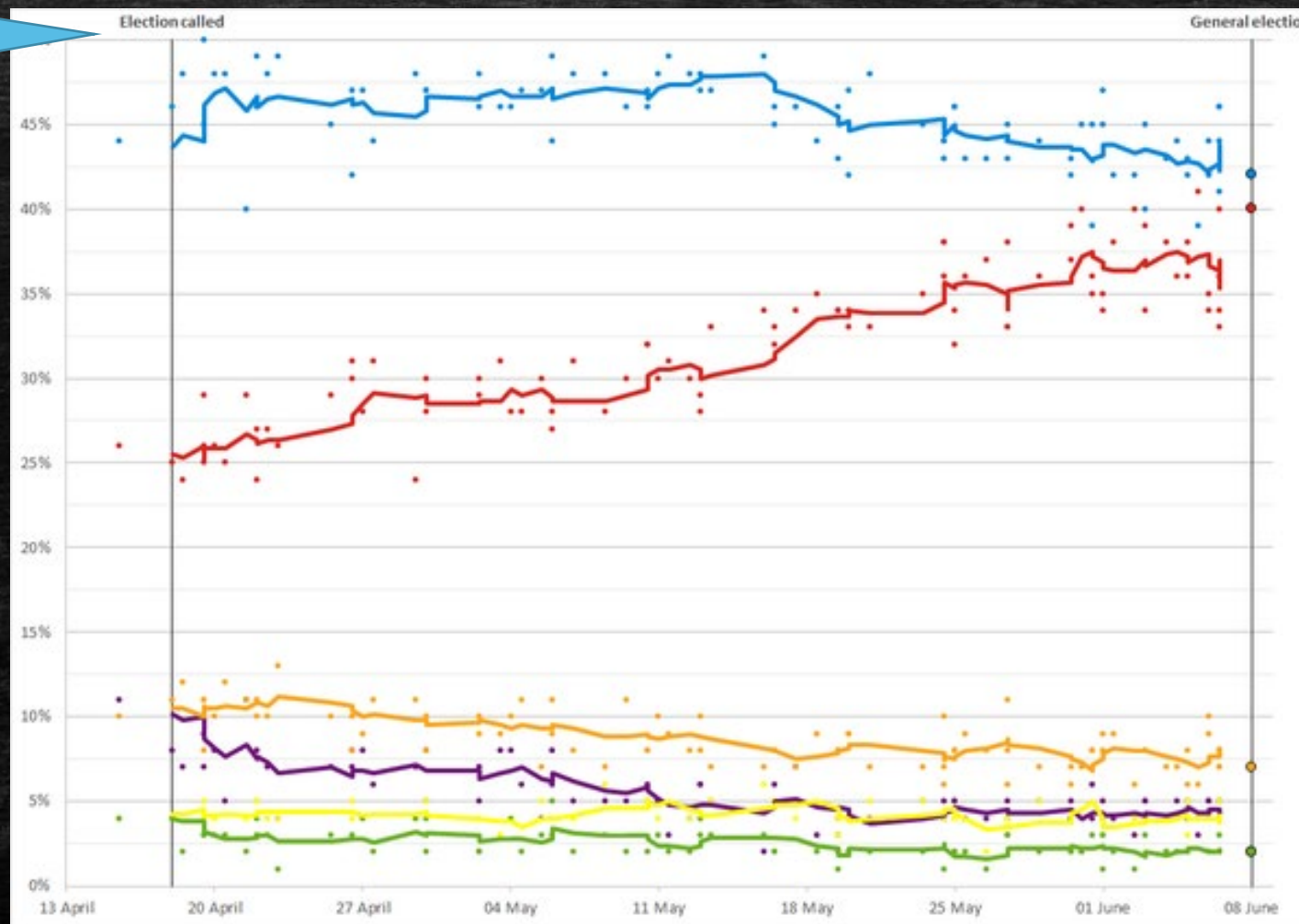
Last  
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# UK Snap Election 2017

Election  
Called





# Predicting Super Majority (350+ seats)

| Parties           | 2015 election result        | Election Forecast <sup>[8]</sup><br>as of 7 June 2017 | Electoral Calculus <sup>[9]</sup><br>as of 7 June 2017 | Lord Ashcroft <sup>[10]</sup><br>as of 6 June 2017 <sup>[11]</sup> | Elections Etc. <sup>[12]</sup><br>as of 2 June 2017 | New Statesman <sup>[13]</sup><br>as of 6 June 2017 | YouGov <sup>[14]</sup><br>as of 7 June 2017 | Britain Elects <sup>[15]</sup><br>as of 7 June 2017 | Scenari Politici.com <sup>[16]</sup><br>as of 7 June 2017 | Forecast UK <sup>[17]</sup><br>as of 8 June 2017 |
|-------------------|-----------------------------|-------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------|---------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------|
| Conservatives     | 330                         | 371                                                   | 361                                                    | 357                                                                | 360                                                 | 339                                                | 302                                         | 353                                                 | 365                                                       | 344-351                                          |
| Labour Party      | 232                         | 199                                                   | 215                                                    | 222                                                                | 210                                                 | 224                                                | 269                                         | 219                                                 | 208                                                       | 221-230                                          |
| SNP               | 56                          | 50                                                    | 48                                                     | 45                                                                 | 48                                                  | 57                                                 | 44                                          | 46                                                  | 49                                                        | 44-52                                            |
| Liberal Democrats | 8                           | 7                                                     | 4                                                      | 4                                                                  | 9                                                   | 8                                                  | 12                                          | 9                                                   | 6                                                         | 5-7                                              |
| Plaid Cymru       | 3                           | 2                                                     | 3                                                      | 1                                                                  | 3                                                   | —                                                  | 2                                           | 3                                                   | 3                                                         | 2-4                                              |
| Green Party       | 1                           | 1                                                     | 1                                                      | 0                                                                  | 1                                                   | —                                                  | 1                                           | 1                                                   | 1                                                         | 0-2                                              |
| UKIP              | 1                           | 1                                                     | 0                                                      | 0                                                                  | 0                                                   | —                                                  | 0                                           | 0                                                   | 0                                                         | 0                                                |
| Others            | 19                          | 1 <sup>[d]</sup>                                      | 18 <sup>[e]</sup>                                      | 19                                                                 | —                                                   | —                                                  | 20                                          | 19                                                  | 18                                                        | 19                                               |
| Overall result    | Conservative majority of 10 | Conservative majority of 92                           | Conservative majority of 72                            | Conservative majority of 64                                        | Conservative majority of 70                         | Conservative majority of 28                        | Hung Parliament<br>(Con 24 seats short)     | Conservative majority of 56                         | Conservative majority of 84                               | Conservative majority of 46                      |



# Except for YouGov

| Parties           | 2015 election result        | Election Forecast <sup>[8]</sup> as of 7 June 2017 | Electoral Calculus <sup>[9]</sup> as of 7 June 2017 | Lord Ashcroft <sup>[10]</sup> as of 6 June 2017 <sup>[11]</sup> | Elections Etc. <sup>[12]</sup> as of 2 June 2017 | New Statesman <sup>[13]</sup> as of 6 June 2017 | YouGov <sup>[14]</sup> as of 7 June 2017 | Britain Elects <sup>[15]</sup> as of 7 June 2017 | Scenari Politici.com <sup>[16]</sup> as of 7 June 2017 | Forecast UK <sup>[17]</sup> as of 8 June 2017 |
|-------------------|-----------------------------|----------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------|------------------------------------------|--------------------------------------------------|--------------------------------------------------------|-----------------------------------------------|
| Conservatives     | 330                         | 371                                                | 361                                                 | 357                                                             | 360                                              | 339                                             | 302                                      | 353                                              | 365                                                    | 344-351                                       |
| Labour Party      | 232                         | 199                                                | 215                                                 | 222                                                             | 210                                              | 224                                             | 269                                      | 219                                              | 208                                                    | 221-230                                       |
| SNP               | 56                          | 50                                                 | 48                                                  | 45                                                              | 48                                               | 57                                              | 44                                       | 46                                               | 49                                                     | 44-52                                         |
| Liberal Democrats | 8                           | 7                                                  | 4                                                   | 4                                                               | 9                                                | 8                                               | 12                                       | 9                                                | 6                                                      | 5-7                                           |
| Plaid Cymru       | 3                           | 2                                                  | 3                                                   | 1                                                               | 3                                                | —                                               | 2                                        | 3                                                | 3                                                      | 2-4                                           |
| Green Party       | 1                           | 1                                                  | 1                                                   | 0                                                               | 1                                                | —                                               | 1                                        | 1                                                | 1                                                      | 0-2                                           |
| UKIP              | 1                           | 1                                                  | 0                                                   | 0                                                               | 0                                                | —                                               | 0                                        | 0                                                | 0                                                      | 0                                             |
| Others            | 19                          | 1 <sup>[d]</sup>                                   | 18 <sup>[e]</sup>                                   | 19                                                              | —                                                | —                                               | 20                                       | 19                                               | 18                                                     | 19                                            |
| Overall result    | Conservative majority of 10 | Conservative majority of 92                        | Conservative majority of 72                         | Conservative majority of 64                                     | Conservative majority of 70                      | Conservative majority of 28                     | Hung Parliament<br>(Con 24 seats short)  | Conservative majority of 56                      | Conservative majority of 84                            | Conservative majority of 46                   |



# Except for YouGov

| Parties           | 2015 election result        | Election Forecast <sup>[8]</sup><br>as of 7 June 2017 | Electoral Calculus <sup>[9]</sup><br>as of 7 June 2017 | Lord Ashcroft <sup>[10]</sup><br>as of 6 June 2017 <sup>[11]</sup> | Elections Etc. <sup>[12]</sup><br>as of 2 June 2017 | New Statesman <sup>[13]</sup><br>as of 6 June 2017 | YouGov <sup>[14]</sup><br>as of 7 June 2017 | Britain Elects <sup>[15]</sup><br>as of 7 June 2017 | Scenari Politici.com <sup>[16]</sup><br>as of 7 June 2017 | Forecast UK <sup>[17]</sup><br>as of 8 June 2017 |
|-------------------|-----------------------------|-------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------|---------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------|
| Conservatives     | 330                         | 371                                                   | 361                                                    | 357                                                                | 360                                                 | 339                                                | 302                                         | 353                                                 | 365                                                       | 344-351                                          |
| Labour Party      | 232                         | 199                                                   | 215                                                    | 222                                                                | 210                                                 | 224                                                | 269                                         | 219                                                 | 208                                                       | 221-230                                          |
| SNP               | 56                          | 50                                                    | 48                                                     | 45                                                                 | 48                                                  | 57                                                 | 44                                          | 46                                                  | 49                                                        | 44-52                                            |
| Liberal Democrats | 8                           | 7                                                     | 4                                                      | 4                                                                  | 9                                                   | 8                                                  | 12                                          | 9                                                   | 6                                                         | 5-7                                              |
| Plaid Cymru       | 3                           | 2                                                     | 3                                                      | 1                                                                  | 3                                                   | —                                                  | 2                                           | 3                                                   | 3                                                         | 2-4                                              |
| Green Party       | 1                           | 1                                                     | 1                                                      | 0                                                                  | 1                                                   | —                                                  | 1                                           | 1                                                   | 1                                                         | 0-2                                              |
| UKIP              | 1                           | 1                                                     | 0                                                      | 0                                                                  | 0                                                   | —                                                  | 0                                           | 0                                                   | 0                                                         | 0                                                |
| Others            | 19                          | 1 <sup>[d]</sup>                                      | 18 <sup>[e]</sup>                                      | 19                                                                 | —                                                   | —                                                  | 20                                          | 19                                                  | 18                                                        | 19                                               |
| Overall result    | Conservative majority of 10 | Conservative majority of 92                           | Conservative majority of 72                            | Conservative majority of 64                                        | Conservative majority of 70                         | Conservative majority of 28                        | Hung Parliament<br>(Con 24 seats short)     | Conservative majority of 56                         | Conservative majority of 84                               | Conservative majority of 46                      |



# And YouGov was right!

| Parties           | 2015 election result        | Election Forecast <sup>[8]</sup> as of 7 June 2017 | Electoral Calculus <sup>[9]</sup> as of 7 June 2017 | Lord Ashcroft <sup>[10]</sup> as of 6 June 2017 <sup>[11]</sup> | Elections Etc. <sup>[12]</sup> as of 2 June 2017 | New Statesman <sup>[13]</sup> as of 6 June 2017 | YouGov <sup>[14]</sup> as of 7 June 2017 | Britain Elects <sup>[15]</sup> as of 7 June 2017 | Scenari Politici.com <sup>[16]</sup> as of 7 June 2017 | Forecast UK <sup>[17]</sup> as of 8 June 2017 | BBC/ITV/Sky exit poll 8 June 2017    | 2017 election result                |
|-------------------|-----------------------------|----------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------|------------------------------------------|--------------------------------------------------|--------------------------------------------------------|-----------------------------------------------|--------------------------------------|-------------------------------------|
| Conservatives     | 330                         | 371                                                | 361                                                 | 357                                                             | 360                                              | 339                                             | 302                                      | 353                                              | 365                                                    | 344-351                                       | 314                                  | 318                                 |
| Labour Party      | 232                         | 199                                                | 215                                                 | 222                                                             | 210                                              | 224                                             | 269                                      | 219                                              | 208                                                    | 221-230                                       | 266                                  | 262                                 |
| SNP               | 56                          | 50                                                 | 48                                                  | 45                                                              | 48                                               | 57                                              | 44                                       | 46                                               | 49                                                     | 44-52                                         | 34                                   | 35                                  |
| Liberal Democrats | 8                           | 7                                                  | 4                                                   | 4                                                               | 9                                                | 8                                               | 12                                       | 9                                                | 6                                                      | 5-7                                           | 14                                   | 12                                  |
| Plaid Cymru       | 3                           | 2                                                  | 3                                                   | 1                                                               | 3                                                | —                                               | 2                                        | 3                                                | 3                                                      | 2-4                                           | 3                                    | 4                                   |
| Green Party       | 1                           | 1                                                  | 1                                                   | 0                                                               | 1                                                | —                                               | 1                                        | 1                                                | 1                                                      | 0-2                                           | 1                                    | 1                                   |
| UKIP              | 1                           | 1                                                  | 0                                                   | 0                                                               | 0                                                | —                                               | 0                                        | 0                                                | 0                                                      | 0                                             | 0                                    | 0                                   |
| Others            | 19                          | 1 <sup>[d]</sup>                                   | 18 <sup>[e]</sup>                                   | 19                                                              | —                                                | —                                               | 20                                       | 19                                               | 18                                                     | 19                                            | 18                                   | 18                                  |
| Overall result    | Conservative majority of 10 | Conservative majority of 92                        | Conservative majority of 72                         | Conservative majority of 64                                     | Conservative majority of 70                      | Conservative majority of 28                     | Hung Parliament (Con 24 seats short)     | Conservative majority of 56                      | Conservative majority of 84                            | Conservative majority of 46                   | Hung Parliament (Con 12 seats short) | Hung Parliament (Con 8 seats short) |



# What did YouGov do?

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- The modeled "every constituency and key voter types in Britain
  - based on analysis of key demographics as well as
    - past voting behaviour",
  - with new interviews to registered voters conducted every day.



# What did everyone else do?

- They adjusted their average based on past demographics.
  - Did not model on a county by county level

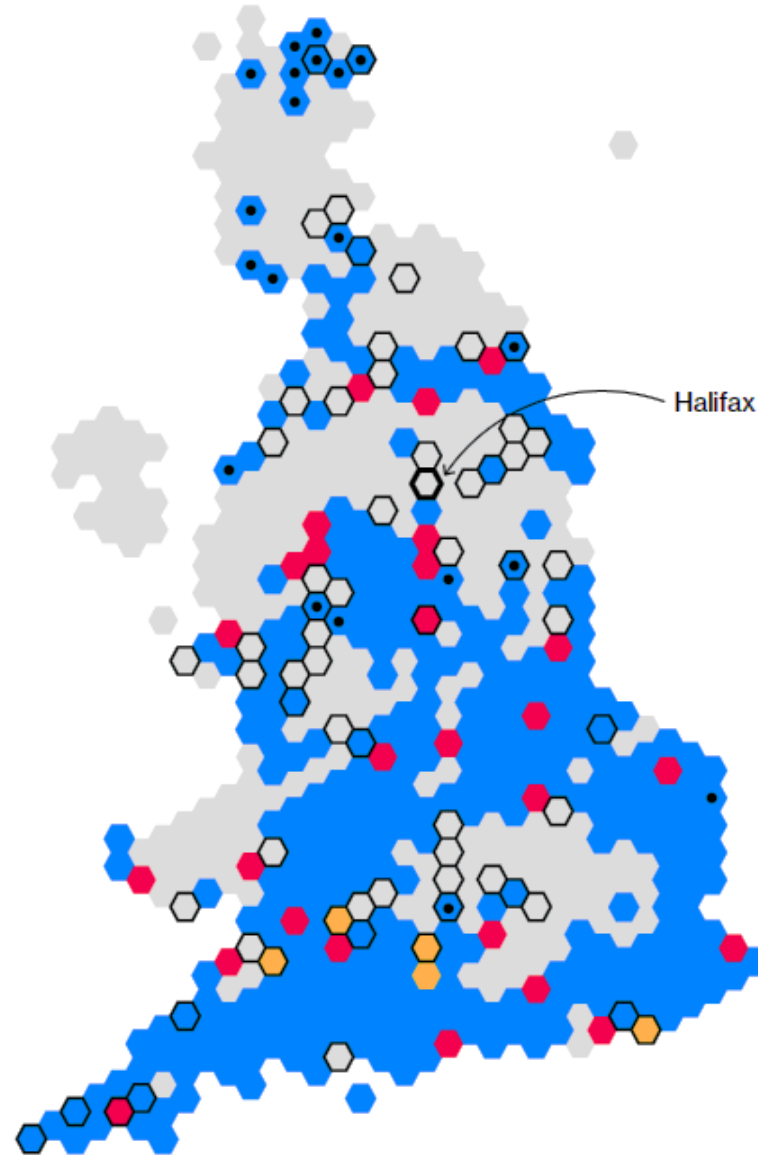
## Pollster adjustments are shifting results toward Conservatives

| POLLSTER       | RAW VOTING INTENTION |             |                                                                                       | HEADLINE VOTING INTENTION |             |                                                                                       |
|----------------|----------------------|-------------|---------------------------------------------------------------------------------------|---------------------------|-------------|---------------------------------------------------------------------------------------|
| Survation      | Tied                 | —           |                                                                                       | Con.                      | +1          |    |
| ICM            | Con.                 | +2          |    | Con.                      | +11         |    |
| ComRes         | Con.                 | +4          |    | Con.                      | +12         |    |
| Opinium        | Con.                 | +4          |    | Con.                      | +6          |    |
| Ipsos MORI     | Lab.                 | +3          |  | Con.                      | +5          |  |
| YouGov         | Con.                 | +1          |  | Con.                      | +3          |  |
| Kantar Public  | Con.                 | +8          |  | Con.                      | +10         |  |
| <b>Average</b> | <b>Con.</b>          | <b>+2.3</b> |  | <b>Con.</b>               | <b>+6.9</b> |  |

SOURCE: HUFFINGTON POST POLLSTER, IPSOS MORI



○ May visited    ● Conservative win    ● Conservative gain    Lost seat to    ● Labour    ● Lib Dem



Sources: Bloomberg reporting, BBC



# Wrapping Up...

